

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			Indicative content: Initially, only weight acts, there is a resultant force downwards and the skydiver accelerates described by Newton's second law, $F = ma$. As the skydiver speeds up air resistance increases until the forces balance. Newton's first law states that if the forces acting on a body are balanced then it will remain in a constant state of motion, so the skydiver travels at terminal speed. Newton's third law states that if a body A exerts a force on body B, then body B exerts an equal and opposite force on body A. As the skydiver exerts a force on the air, the air exerts an equal and opposite force on the skydiver which is air resistance. 5–6 marks Correctly states all 3 laws and correctly applies them to the motion. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks Correctly states and applies 2 laws or does 3 partially. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks Correctly states and applies 1 law or a partial treatment of 1 or 2. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	3	3		6		

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	(b)	(i)		Gradient = $\frac{(54 - 13)(1)}{(9.5 - 0.5)(1)}$ Gradient = $\frac{41}{9} = 4.6 \text{ [m/s}^2\text{]} (1)$ Accept range $4.6 \pm 0.3 \text{ [m/s}^2\text{]}$		3		3	3	
		(ii)		[Acceleration] decreases (1) to zero [at 15 s] (1)		2		2	1	
		(iii)		Distance travelled = area under line stated or implied or $x = \frac{1}{2}(u + v)t \text{ (1)}$ $(\frac{1}{2} \times 5 \times 33) (1)$ $= 82.5 \text{ [m]} (1)$ Accept answers in the range 80-85 [m]	1	1 1		3	3	
				Question 7 total	4	0	0	14	7	0